
Formalizing AI Scientist Systems: A Component Framework and Theoretical Analysis

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Abstract

The field of *AI Scientist* systems—agents that generate hypotheses, execute experiments, and synthesize knowledge—lacks a shared formal vocabulary for cross-system comparison. We propose Servo (S, G, E, V, M, π), a six-component taxonomy comprising the Search space, hypothesis Generator, Experiment executor, Validator, Memory, and search Policy (π), grounded in Partially Observable Markov Decision Process (POMDP) and Bayesian Experimental Design (BED) theory. The framework’s core contribution is three diagnostic separations that current systems conflate: the observation likelihood from the validator as a reward channel; calibrated gating (V_{gating}) from mere layer presence (V_{present}); and episodic from semantic memory. We formalize these separations as three propositions under tractable, identifying assumptions—the discovery loop closes independently of the validator; a miscoupled validator drives convergence to the wrong target; and a novelty component absent from every automated channel is non-identifiable (Propositions 1–3)—noting that no surveyed system satisfies those assumptions, so the propositions clarify the formal structure of the distinctions rather than certify their empirical prevalence. We apply the vocabulary to six core AI Scientist systems and seven scientific domains, and identify three open problems with falsification criteria: the lack of validated automated novelty gates, the BED–practice gap for search policies, and the underdevelopment of semantic memory M_s . The cross-system survey is offered as provisional structured annotation; the organizing hypothesis that gating calibration co-occurs with trustworthy closure is direction-consistent under some outcome-recoding assumptions but not adjudicated, and should not be read as an empirical finding.

1 Introduction

AI Scientist systems—agents that independently generate hypotheses, execute experiments, and synthesize knowledge—have proliferated rapidly [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025, Feng et al., 2026, Liu et al., 2026], yet the field lacks a shared formal vocabulary. Systems range from tool-augmented synthesis agents (Coscientist [Boiko et al., 2023]) through closed-loop manuscript-writing pipelines (The AI Scientist [Lu et al., 2024, 2026]) to formal-mathematics agents (Aletheia [Feng et al., 2026]), each introducing bespoke terminology that makes systematic cross-system comparison difficult. Without a shared decomposition, questions that bear directly on deployment remain unanswered: *What determines whether a system can operate in a closed loop? Why does autonomous novelty evaluation fail consistently? What is the theoretically optimal search policy?*

*Prepared for the Conference for AI Scientists (CAISc) 2026. Fitting the venue, the work was itself AI-assisted: drafting and analysis used Claude Code (Anthropic), and the cross-system reliability coding used three model-pinned LLM agents from different vendors (Anthropic, OpenAI, Google; §8 and supplement). The author provided direction, design, source verification, and final judgment; see the AI Involvement Checklist.

We propose Servo² (S, G, E, V, M, π) , a formal framework grounding AI Scientist systems in POMDP and Bayesian Experimental Design theory (Section 4). We apply it to analyze four core systems, an artifact infrastructure proposal, and instantiations across seven scientific fields (Sections 5, 6); and identify three open problems with falsification criteria (Section 7). The analysis reveals a recurring diagnostic distinction: systems that mechanically close the loop differ sharply in whether their acceptance gate, novelty judgment, and memory structure can support trustworthy discovery claims; we develop the supporting theory in Appendix A. The contribution is thus a diagnostic vocabulary that keeps mechanical closure, validator calibration, novelty judgment, and memory separable rather than collapsing them into a single notion of autonomy.

Scope. We include systems whose primary goal is automated scientific *discovery* through hypothesis-experiment-validation cycles, and exclude specialized predictors without a generation-and-search loop (e.g., AlphaFold [Jumper et al., 2021]). Within that scope we analyze widely-cited representative systems rather than an exhaustive enumeration; this convenience sampling introduces selection bias that we flag in Section 8.

2 Background

Following Kaelbling et al. [1998], we model AI Scientist systems as *partially observable Markov decision processes*: the true state s (unknown structure of nature) is inaccessible; the system maintains a belief $b(s)$ updated by experimental observations. Formally, an AI Scientist POMDP is the tuple $(S, \mathcal{A}, T, R, \Omega, \mathcal{O}, b_0, \gamma)$, where $\mathcal{A}$ is the action space (hypothesis generation and experiment execution), T is the transition function, R is the reward (scientific value of discovery), $\mathcal{O}$ is the noisy observation process, b_0 is the initial belief, and γ the discount factor; the belief is updated by Bayes’ rule, $b'(s') \propto \mathcal{O}(o | s', a) \sum_s T(s' | s, a) b(s)$, and π acts on the belief b rather than the inaccessible state s . Three properties distinguish this from standard control: R cannot be pre-specified (significance is not defined a priori); S is unbounded; and T is non-stationary (accumulated knowledge changes the exploration frontier). We therefore use the POMDP as an organizing formalism rather than a fully specified model of open-ended science: the formal results of Appendix A are established for the tractable subproblem (Assumption 1) in which these departures are bounded, while the general open-ended case is left open. No surveyed system satisfies Assumption 1 in full—all operate in regimes where S is unbounded, T is non-stationary, or R cannot be pre-specified—so the propositions apply to an idealized subproblem rather than directly to any system analyzed here; their contribution is to clarify the formal structure of the distinctions the survey draws, not to certify that those distinctions hold in practice. Throughout, three distinct senses of “loop closure” should be kept separate: (i) posterior concentration on θ^* in the tractable subproblem (Propositions 1–3); (ii) *computational* closure, where feedback from V flows back into G ; and (iii) *trustworthy* closure, where the acceptance gate reliably tracks truth. The propositions establish (i); the survey concerns (iii); systems may achieve (ii) without (i) or (iii). Bayesian Experimental Design [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] provides a normative criterion for the information objective of the search policy π : select d^* maximizing Expected Information Gain (EIG),

$$d^* = \arg \max_{d \in \mathcal{D}} \text{EIG}(d) = \arg \max_d \mathbb{E}[H(\theta) - H(\theta | y, d)] \quad (1)$$

where $H(\theta)$ is prior entropy over parameters θ and $H(\theta | y, d)$ is expected posterior entropy after observing y under d . EIG can be approximated via variational methods [Foster et al., 2019], amortized policies [Foster et al., 2021], and RL [Blau et al., 2022]. Domain-specific autonomous-science systems already use active-learning or uncertainty-guided approximations—the Robot Scientist’s experiment selection [Sparkes et al., 2010] and GNoME’s DFT-in-the-loop active learning [Merchant et al., 2023]—but recent open-ended LLM-based systems maintain no explicit posterior over hypotheses or stated EIG objective; that narrower gap is the one we identify in Section 7.

²We name the framework after the *servomechanism*: as a servo loop converges on a target by acting on a feedback (error) signal, an AI Scientist closes the discovery loop only when its validator V supplies a reliable feedback signal—the framework’s central concern.

3 Related Work

AI Scientist systems. Prior work [Lu et al., 2024, 2026, Schmidgall et al., 2025, Boiko et al., 2023] provides qualitative characterizations of individual systems but no shared framework for cross-system comparison; idea-generation studies [Si et al., 2024, Baek et al., 2024] establish quality baselines without unifying the full discovery loop; knowledge-graph-driven ideation systems [Gu and Krenn, 2024, Lee et al., 2025] further show that structuring semantic memory as a relational corpus can substantially improve hypothesis-generation quality, but equally without closing the loop.

Autonomous-science frameworks. Closed-loop autonomous discovery predates the LLM era: the Robot Scientist [Sparkes et al., 2010] integrated hypothesis generation, robotic experimentation, and statistical validation in yeast functional genomics, instantiating all six Servo components—including an active-learning policy—in physical wet-lab form more than a decade before LLM-based systems. More recent unified frameworks include NovelSeek [Zhang et al., 2025], a closed-loop multi-agent system spanning idea generation to verification across twelve scientific tasks, and Co-Scientist [Gottweis et al., 2026], a multi-agent hypothesis-generation system with experimental biomedical validation. We therefore position Servo not as the first unified framework for automated discovery, but as a formal vocabulary that makes validator completeness, semantic memory, and the POMDP/BED relationship explicit within the recent LLM-agent subfield. The incremental diagnostic value over existing frameworks is most visible at two points. First, standard POMDP classification and NovelSeek’s multi-agent framework distinguish loop-closed from loop-open but do not separate V_{present} (a calibrated layer exists) from V_{gating} (the decisive acceptance gate is calibrated); AI Scientist (Nature 2026) illustrates this: both NovelSeek’s vocabulary and a plain POMDP characterization code it as closed-loop, but Servo’s V-layer decomposition locates the failure—the gating layer is biased, not absent—which standard vocabulary cannot express. Second, existing POMDP treatments collapse loop closure into a binary, while Servo’s three-way separation (posterior concentration, computational closure, trustworthy closure) identifies systems such as Agent Laboratory that close a computational feedback loop without trustworthy gating, a distinction that would otherwise conflate mechanical and trustworthy closure.

Decision-theoretic foundations. The POMDP framework [Kaelbling et al., 1998] and BED [Chaloner and Verdinelli, 1995, Rainforth et al., 2024] are classical. Bayesian-optimal experiment selection has a long history in automated science—most directly in the Robot Scientist’s active-learning policy [Sparkes et al., 2010]—and we build on this tradition, connecting POMDP (structural formulation) to BED (normative policy criterion) for the modern LLM-agent setting rather than claiming priority for the idea itself. The reward specification and specification gaming literatures study the consequences of unreliable reward signals; Servo extends this lens to AI Scientist systems by stratifying V into reliability levels and connecting each level to specific closed-loop failure modes.

Formal models and artifact infrastructure. Taniguchi et al. [2025] propose Collective Predictive Coding (CPC-MS) as a social model of science, recasting scientific activity as decentralized Bayesian inference across a community. CPC-MS and Servo are complementary: CPC-MS provides a *collective* and *social* model in which peer review is a Metropolis-Hastings consensus process; Servo provides the *single-system* and *modular* decomposition that must be instantiated at each node. The three open problems we identify are precisely the necessary conditions for a CPC-MS-compliant collective system to be realizable. Liu et al. [2026] address the artifact layer — output format and verification protocol — rather than the discovery loop itself; we position ARA as a V and M_s infrastructure contribution within Servo, discussed in Sections 5.1 and 7.

4 The Servo Framework

We define an AI Scientist system as a tuple (S, G, E, V, M, π) with the following components:

$$\text{Servo} = (S, G, E, V, M, \pi) \quad (2)$$

S : Search Space The set of candidate hypotheses, programs, molecules, experimental conditions, or other objects subject to exploration. S may be static or dynamically expanding (open-ended).

$G : M \times S \rightarrow h$: **Hypothesis Generator** A function that proposes new candidate $h \in S$ conditioned on accumulated memory M and the current state of the search space. G may involve LLM generation, evolutionary mutation, or symbolic search.

$E : h \times P \rightarrow o$: **Experiment Executor** A function that executes hypothesis h under protocol P to produce observation o . The fidelity of E ranges from fast computational simulation (low cost, potential surrogate error) to physical wet-lab execution (high cost, ground truth).

$V : o \rightarrow r$: **Validator** A function that assigns a reward signal r to observation o . We distinguish four layers of V :

$V_{\text{syntax}} : o \rightarrow \{\text{pass, fail}\}$	(deterministic, fast)
$V_{\text{semantic}} : o \rightarrow r \in [0, 1]$	(LLM-based, stochastic)
$V_{\text{empirical}} : o \rightarrow r$	(reproduction-based)
$V_{\text{human}} : o \rightarrow \{\text{novel, significant}\}$	(human-delegated)

M : **Memory** Structured as (M_e, M_s, M_p) . M_e (episodic) stores individual records (h_i, o_i, r_i) ; M_s (semantic) stores distilled knowledge patterns; M_p (procedural) stores learned protocols. The structure of M directly determines the quality of G and π .

$\pi : b(S) \times M \rightarrow h$: **Search Policy** The decision rule for selecting the next hypothesis. The belief state $b(S)$ is updated after each experiment. In the BED interpretation, $\pi^* = \arg \max_d \text{EIG}(d)$ (Equation 1). In practice, current systems use policies ranging from reactive (ReAct) to greedy to tree search.

The operational loop is:

$$\begin{aligned} \pi(b, M) &\rightarrow G(M_s, S) \rightarrow E(h, P) \rightarrow V(o) \\ &\rightarrow M += (h, o, r) \rightarrow b(S) \text{ update} \rightarrow \text{repeat} \end{aligned} \quad (3)$$

We further introduce $H \in [0, 1]$ (distinct from the entropy function $H(\theta)$ in Eq. 1) as an auxiliary parameter denoting the degree of human intervention: $H = 0$ for fully autonomous systems and $H = 1$ for systems in which humans assume the role of G or π .

V-completeness as a design dimension. The layers form a partial order under reliability: V_{syntax} deterministic; V_{semantic} stochastic and bias-prone; $V_{\text{empirical}}$ from well-calibrated (DFT) to poorly calibrated (GNN surrogates); V_{human} irreplaceable for novelty but rate-limiting; and V_{formal} (proof kernels, type-checkers) the mechanically-decidable ideal limit, used for mathematics (formal) in Table 2. We define V *completeness* as the highest layer both accessible and reliably calibrated; accessibility without calibration does not count (Agent Laboratory’s biased reviewer yields only imperfect completeness). We treat it as a central design variable because it caps what π can learn and what G can be rewarded for: without a reliable V , no policy can distinguish progress from regression. What matters is not that a calibrated layer is *present* but that the layer which *gates acceptance* is calibrated: a calibrated check can be neutralized when a biased semantic or social layer decides acceptance (§8). The cross-domain pattern (Section 6) appears to track the calibration of the gating validator more than G or π sophistication; we develop the relevant formal results next.

The validator as a reward channel. Treating V as distinct from the observation likelihood, we prove three results in a tractable, identifying subproblem (Appendix A): under identifiability and divergent accumulated information the loop closes (Proposition 1); a validator miscoupled into the belief update drives convergence to the wrong target (Proposition 2); and a novelty component invisible to every automated channel is non-identifiable (Proposition 3). A complete validator is thus *necessary* for trustworthy autonomous closure—i.e., for truth-tracking reward assignment—but, by Proposition 1, not sufficient without enough accumulated information; the surveyed systems illustrate these results rather than prove them.

5 Analysis of Core AI Scientist Systems

We apply Servo to four end-to-end systems [Boiko et al., 2023, Lu et al., 2024, 2026, Schmidgall et al., 2025] and one artifact infrastructure proposal. Table 1 synthesizes the key dimensions; in this small sample, the systems that reach closed-loop operation are also those with more complete validators, although they also advance G , E , and π (we revisit this interaction in Section 8).

	Robot Sci.	Coscientist	AI Sci. (2024)	AI Sci. (2026)	AgentLab	NovelSeek
Novelty measure	Stat. test	None	LLM self-score	3-layer	Human-deleg.	Perf. gain
Loop closed?	Yes (wet-lab)	Partial	Yes (greedy)	Yes (tree)	Partial	Yes (comp.)
π type	Active learn.	ReAct	Greedy	Tree search	Human/seq.	Feedback
V completeness	V_e	$V_h + V_e$	V_s (biased)	$V_e + V_s + V_h$	V_s (biased)	V_e
M structure	$M_e + M_s$	Ephemeral	M_e	$M_e + M_s$	M_e	M_e
H (approx.)	Low	$G (\approx 1)$	Low (≈ 0)	Low (≈ 0)	$G + \pi (\approx 0.8)$	Low

Table 1: Comparative Servo analysis of core AI Scientist systems, including the pre-LLM Robot Scientist [Sparkes et al., 2010] and the recent unified framework NovelSeek [Zhang et al., 2025]. V_s =semantic, V_e =empirical, V_h =human. For AI Scientist (Nature 2026), V_s is present but non-gating; the decisive acceptance gate is V_h (ICLR peer review), consistent with $V_gating=0$ in the released coding (analysis/systems.csv).

Across the four systems, the ones that close the loop are also those that upgrade or add validators; this does not mean G or π advances are absent—several systems also upgrade ideation, tool use, and search—but in these cases validator reliability is the constraint that gates whether such advances can be exploited in a closed loop. Coscientist has only constrained task-level feedback (measured outcomes inform later choices [Boiko et al., 2023]); its loop is *partial*, with novelty/significance validation human-mediated. The AI Scientist (Sakana) adds V_s via a 5-ensemble GPT-4o reviewer (balanced accuracy 0.65 vs. human 0.66 [Lu et al., 2024]), enabling the first *structurally* closed loop on a biased, uncalibrated V . The AI Scientist (Nature) stacks V_e (replication nodes with mean $\pm$ s.d.), V_s (automated reviewer, balanced accuracy 0.69 [Lu et al., 2026]), and actual peer review (V_h via ICLR workshop), giving the most layers in the class—though its automated acceptance gate (the biased V_s reviewer) stays uncalibrated (§8). Agent Laboratory delegates V to V_s (NeurIPS-guideline LLM reviewer) but documents +2.3-point systematic over-estimation relative to human PhD students—illustrating that adding a V layer does not guarantee V completeness if the layer is unreliable [Kim et al., 2026]. Within this sample, no G or π advance improved loop feasibility without a corresponding V improvement—though four systems cannot establish a general impossibility.

5.1 Artifact Infrastructure: Agent-Native Research Artifacts

Liu et al. [2026] address a complementary problem—what the *output* of the loop should be and how to verify it. Agent-Native Research Artifacts (ARA) replace the narrative paper with a four-layer ontology (/logic, /src, /trace, /evidence); the ARA Seal automates V_{syntax} , V_{semantic} (Rigor Auditor, 82.6% detection), and $V_{\text{empirical}}$ (sandboxed reproduction), reserving V_{human} for significance and novelty [Liu et al., 2026]. Motivating figures: only 45.4% of papers fully specify reproduction requirements and 90.2% of extension cost is failed exploration. The /trace layer externalizes M_e/M_s , but preserved traces both accelerated and anchored downstream agents—richer M_e does not monotonically improve π .

6 Domain Analysis

Table 2 is an illustrative domain map: it uses the Servo vocabulary to expose how validation bottlenecks differ across seven scientific fields, the bottleneck following from each field’s V type (Propositions 1–3; per-domain analysis with sources in Appendix B). Systems analyzed span mathematics (formal: [Romera-Paredes et al., 2024, Xin et al., 2024]; NL: [Feng et al., 2026]), physics [Udrescu and Tegmark, 2020, Cranmer et al., 2020], chemistry [Bran et al., 2024, Sprueill et al., 2024], materials [Merchant et al., 2023], biology [O’Donoghue et al., 2023], social science [Manning et al., 2024, Aher et al., 2023, Park et al., 2023a], and CS/algorithms [Real et al., 2020, Sartori and Blum, 2024, Jaber and Jaber, 2026].

The pattern is largely shaped by V type. Formal mathematics has a binary V_{formal} that makes RL stable and the loop fully closed; natural-language proof reintroduces V_{human} (Aletheia: 6.5% audited-correct). Chemistry and biology close the computational loop but leave the physical loop open at the measurement boundary, where characterization stays human-delegated. Materials science is the surveyed domain whose policy most closely resembles active experimental design within a tractable computational surrogate: GNoME uses model uncertainty and DFT relaxations to close a computational loop for phase-stability prediction. DFT approximates physical energies rather than constitut-

Domain	V type	V reliability	Loop	π	Primary bottleneck
Math (formal)	V_{formal}	Perfect oracle	Fully closed	RL/MCTS	Pretraining contamination
Math (NL)	$V_{\text{sem}} + V_h$	Imperfect	Closed (low quality)	Seq. refine	Hallucination; plagiarism
Physics	$V_{\text{emp}} + V_h$	Medium	None/partial	Deterministic	No cross-problem M
Chemistry	V_{emp} (surr.)	Medium	Comp. only	Beam	Wet-lab loop open
Materials	V_{emp} (DFT)	High (comp.)	Comp. closed	Uncert. samp.	DFT wall-time; synth. lag
Biology	$V_{\text{emp}} + V_h$	Low–med	Partial	None	V metrics insufficient
Social Sci.	Stat. test	Low–med	Partial	Factorial	Circular validity
CS / Algo.	V_{emp}	High	Fully closed	Evol./LLM	Interpretability absent

Table 2: Illustrative Servo domain mapping across seven fields (per-domain coding in `domain_systems.csv`). V reliability is an ordinal calibration label, not measured agreement with ground truth. Domain codings are single-coder interpretations of the cited primary sources; no inter-coder κ statistics are available for this table.

ing a fully calibrated empirical oracle, and most predicted stable compounds still require synthesis, characterization, and functional validation [Cheetham and Seshadri, 2024]; the physical materials-discovery loop therefore remains only partly closed. Social science suffers circular validity when G and E share a model [Aher et al., 2023], and CS/algorithms close the loop via reliable V_{emp} but lack interpretability.

7 Open Problems

Our analysis identifies three open problems that persist across all systems and domains. For each we state a *proposed* falsification test—a concrete design whose outcome would refute the corresponding claim—while noting that none are executed here; they specify what evidence would settle the question rather than constituting completed experiments.

7.1 The Lack of Validated Automated Novelty Gates

No current automated validator reliably evaluates novelty: every system from Coscientist (no mechanism) to The AI Scientist Nature (three-layer validation) converges on this conclusion. Two failure modes are documented: LLM self-evaluation bias (systematic over-estimation; hallucinated results) and pretraining contamination (systems may “discover” results implicit in training data, as Aletheia documents for Erdős problems). The ARA Seal’s Rigor Auditor [Liu et al., 2026] achieves 82.6% average detection across five structural violation types but only 22% on orphaned-experiment violations (different denominators); it cannot cross the novelty boundary. Concretely: Agent Laboratory’s LLM reviewer over-estimated paper quality by +2.3 points on the NeurIPS review scale relative to human PhD students; The AI Scientist (Sakana) hallucinated ablation tables; and of the 200 Erdős solutions selected for human audit (from 700 attempted), only 13 (6.5%) were rated meaningfully correct. A 45-expert evaluation of human and AI reviews of Nature-family papers likewise positions current AI reviewers as complements to, not substitutes for, human experts—strong on aggregate criticism quality yet sharing recurring blind spots and overlapping far more than independent humans do [Kim et al., 2026]. The Leiden Declaration [Alper et al., 2026], endorsed by the International Mathematical Union and signed by over 130 mathematicians, independently corroborates the indispensability of expert oversight in mathematical research: AI-produced proofs may contain almost-invisible errors that resist detection without human scrutiny, reinforcing that no automated validation layer fully substitutes for V_{human} even in the domain where automated checking is most mature.

Proposed falsification test (design; not yet executed): The thesis is refuted if an automated validator matches a panel of ≥ 3 domain experts on novelty discrimination—Cohen’s $\kappa \geq 0.80$ against the expert majority—on a benchmark of published-versus-withheld findings whose training-set membership is fixed by a pretraining cutoff date, across ≥ 3 distinct domains. Constructing the

contamination-controlled benchmark and the expert baseline is itself an open task; we specify the instrument and threshold but do not run the test.

Novelty, significance, and correctness are distinct properties that current systems collapse into one scalar (e.g., The AI Scientist’s 1–10 score [Lu et al., 2024]); a viable automated novelty V would need a contamination-controlled reference corpus, architectural separation between G and V , and a definition of “novel to the field” beyond lexical training-set membership (e.g. disruption or atypicality measures [Park et al., 2023b, Uzzi et al., 2013])—none of which current systems provide.

7.2 The BED–Practice Gap for π

Equation 1 specifies the theoretically optimal policy, yet no surveyed AI Scientist system implements it: all deployed policies in the systems analyzed here are heuristic (ReAct, greedy, tree search, human-guided). The partial exception is GNoME [Merchant et al., 2023], whose deep-ensemble uncertainty sampling approximates EIG over a tractable GNN posterior—but only because DFT provides a relatively well-calibrated computational V for phase-stability screening and the hypothesis space admits a learnable probabilistic model, conditions absent from the LLM-based AI Scientist systems surveyed here, where the “posterior” is implicit in model weights [Foster et al., 2021]. Recent work (BED-LLM [Choudhury et al., 2026]) derives a usable EIG posterior from an LLM’s predictive distributions, but for conversational queries rather than scientific experiment design—so the gap is integration with reliable E and V , not impossibility. Concretely, every system in Table 1 degrades to heuristic π : Coscientist selects the next tool reactively, The AI Scientist (Sakana) greedily avoids archive similarity, and Agent Laboratory follows a fixed sequential schedule—none improve with additional experimental data, the failure mode EIG-optimal π would avoid.

Proposed falsification test (design; not yet executed): The gap is refuted if a system explicitly computes and maximizes EIG over a posterior $p(\theta \mid y)$ whose hypothesis space admits no closed-form or finite-ensemble surrogate—operationalized as a posterior defined directly over natural-language hypotheses or program text rather than over a hand-specified low-dimensional model—thereby excluding tractable-model proxies such as GNoME’s deep-ensemble policy. We give this boundary as the measurable criterion but do not test it.

Architecturally, EIG needs a posterior $p(\theta \mid y)$, but in LLM systems θ is implicit in frozen weights with no per-experiment update. Either one maintains an explicit probabilistic surrogate (GNoME’s route, viable only where a calibrated input–outcome model exists) or uses amortized EIG estimation [Foster et al., 2021], which remains unsolved for open-ended $\mathcal{S}$.

7.3 The Underdevelopment of Semantic Memory M_s

Current systems store raw experiment records (M_e) but do not distill them into transferable patterns (M_s): a system with 1,000 experiments does not search more intelligently than one with 10. The ARA /trace layer [Liu et al., 2026] is the closest approximation, but traces accelerated extension performance while also anchoring capable agents to prior solution paths— M_s structure must be paired with selective forgetting mechanisms to be beneficial. Concretely, GNoME’s GNN—one of the few systems where $G(M_s, S)$ genuinely improves—accumulates domain-specific thermodynamic knowledge after 460K DFT calculations, yet this knowledge does not transfer to chemistry or biology; the system is no more capable in any domain it has not explicitly trained on.

Knowledge graphs and world models represent one concrete structural form M_s could take. Large-scale knowledge graph infrastructure—such as SciAtlas [Qiao et al., 2026]—provides one structural substrate; Spacer [Lee et al., 2025] demonstrates the G improvement that such structure can yield: knowledge-graph-driven ideation achieves an AUROC of 0.737 in predicting publication impact, with outputs rated closer to top-journal articles than those from SOTA LLMs. The gain is real, but Spacer operates outside a closed loop—absent E and a calibrated V , the improvement cannot be exploited by a policy that distinguishes progress from regression. This is precisely what the open problem names: partial M_s architectures exist, but their integration into a trustworthy closed loop remains unsolved.

Proposed falsification test (design; not yet executed): The bottleneck is refuted if, with the base model held fixed, a system’s G -generated hypothesis quality improves monotonically with the number of stored prior experiments on a held-out task family it was not trained or built for—measured as

expert-rated quality or downstream task success against a pre-registered effect-size threshold, with an ablation that removes M_s . We specify the protocol but do not execute it.

The hard M_s problems arise when memory is lossy, compressed, conflicting, or strategically filtered—each of which can degrade G rather than improve it. GNoME shows within-domain accumulation works, but its knowledge does not transfer across domains; building M_s that generalizes, forgets selectively, and tolerates conflicting evidence is the open precondition for systems whose $G(M_s, S)$ improves with scale.

The three open problems are not independent. A system with complete V and tractable M_s would have the prerequisites for an EIG-optimal π : complete V supplies the calibrated reward signal that reward-guided policy selection needs, and tractable M_s supplies the hypothesis-space model over which information gain can be estimated. Whether they suffice in non-stationary or unbounded S is open, but they are necessary: the problems converge on a single structural gap where progress on one unlocks the others. They are also the necessary conditions for a collective AI co-scientist [Taniguchi et al., 2025]: $V_{\text{collective}}$ needs solved novelty automatability, a principled collective π a closed BED—practice gap, and shared memory w a developed M_s —with ARA [Liu et al., 2026] a plausible entry point that externalizes M_s and calibrates the sub-novelty layers of V .

8 Discussion

Validator design as a constraint. Our analysis suggests V design is a necessary constraint on closed-loop operation: unreliable validation prevents a system from learning whether its outputs represent progress. The surveyed systems do not isolate V from G , E , M , and π , so we read V completeness as a recurring bottleneck, not an established causal priority. The practical implication still stands: before optimizing G or π , designers should ask what level of V completeness the domain actually supports. This design guidance derives from the diagnostic vocabulary’s utility for localizing failure modes—not from the empirical association between calibration and trustworthy closure, which remains a framing hypothesis rather than a validated finding. At the same time, V -completeness and a BED-optimal π are the key architectural requirements for *distinct* goals: trustworthy closure requires a calibrated validator, but scientifically valuable closure—reaching genuinely unexplored regions of S —requires a search policy that escapes the already-exploited frontier. Current systems fall short on both, and the failure modes do not reduce to one another: improving V does not ensure π ventures beyond familiar territory, and improving π without calibrated V produces blind exploration that cannot distinguish progress from regression.

Human oversight. Rather than treating $H \in [0, 1]$ as a quantity to minimize, our analysis suggests explicit human integration at the V_{human} and π levels is both practically necessary and theoretically appropriate: human judgment accesses an information source no automated proxy yet replicates, particularly for the non-automatizable components of novelty and significance. The goal is not full autonomy but *appropriate* autonomy—closing the loop where V is reliable while preserving human judgment where it is not.

Towards collective AI co-scientists. All systems analyzed are single-agent loops; extending Servo to a multi-agent collective with shared memory w and a collective validator [Taniguchi et al., 2025] requires exactly the three open problems above to be solved, with ARA [Liu et al., 2026] a plausible entry point for w . Such a collective must also preserve epistemic diversity: AI reviewers already overlap far more than humans [Kim et al., 2026], risking the loss of the transient diversity that benefits epistemic communities [Zollman, 2010]. The Leiden Declaration [Alper et al., 2026] identifies a compounding risk: as AI systems target problems amenable to automated verification, research may be skewed toward AI-friendly tasks, concentrating the community’s collective search on a narrowing frontier and distorting S at the field level—a failure mode that no single-system V or π improvement can address.

Limitations. Servo abstracts away several factors: the cost structure of E (orders of magnitude across domains), the non-stationarity of S , and the single-agent framing’s social layer [Taniguchi et al., 2025]. More fundamentally, the V -completeness pattern is a cross-system *association*, not an established causal determinant, and the surveyed systems are a convenience sample subject to selection and survivorship bias. Mechanical loop closure moreover does not require completeness (AI Scientist v1 and NovelSeek close a computational loop at low completeness), so the claim concerns *trustworthy* closure, which we do not measure independently of V ; we present it as an organizing

hypothesis, not a tested association; the supplement’s `trustworthy_closure` coding draws on outcome signals only partly outside V (peer review and publication overlap V_{human}) and, at six systems with outcomes entangled with V , cannot test the direction. To make the coding source-linked and formally traceable we release the rubric, coding sheets with a representative source quote and citation per row (core systems and the cross-domain Table 2), and validator scripts that reject any uncited or unquoted row; the scripts verify formal presence of quotes and citation keys but do not validate semantic entailment of each coded field from those passages—per-field judgment still requires consulting the cited primary source. Three independent, model-pinned, blind LLM-based coder agents from different vendors (Anthropic, OpenAI, Google)³ coded 14 Tier-1 systems for reliability estimation (see `multicoder/target_systems.md`); the six core systems in Table 1 are a representative analytical sample drawn from that broader set, selected to span the discovery-loop spectrum. inter-coder agreement (Fleiss’ κ over the 13 jointly-coded systems) is substantial-to-perfect on most constructs (calibration $\kappa=0.79$, loop status $\kappa=0.74$, semantic-validation presence $\kappa=1.0$; mean $\kappa=0.68$) and only fair on the holistic V -completeness ordinal ($\kappa=0.39$); against the author’s reference labels specifically, Cohen’s κ is negative (-0.25 to -0.43) on this ordinal—the construct on which the author’s coding inverts independent coders and, not coincidentally, the construct the paper’s central claim does not rest on (the claim rests on the calibration sub-construct, $\kappa=0.79$). All reported as automated transparency, not human-level reliability. A counterexample search found no V -incomplete system that achieves *trustworthy* closed-loop discovery: the lone near-miss (AI Scientist v1) closes on a biased validator but yields unreliable output. Propositions 1–3 formalize the three separations underlying this vocabulary—observation vs. reward channel, coupled vs. uncoupled validator, and identifiable vs. non-identifiable novelty component—under the tractable subproblem assumptions; they do not establish the empirical V -completeness association. The coders split from those labels—AI-produced under human direction (Coder A in the supplement), not a human gold standard—on whether AI Scientist (Nature 2026) has a “calibrated” validator—the old label conflated a calibrated layer being *present* with the *gating* layer being calibrated. Re-coding the split over all 14 systems (two vendors; supplement) resolves it (the reference labels and both coders agree the layer is present but non-gating); gating calibration is more clearly specified by this two-vendor recoding (raw agreement 13/14, Cohen’s $\kappa=0.86$; an automated robustness check, not expert-validated reliability), clarifying why mere calibrated-layer presence is insufficient, and no system has a calibrated *automated* novelty gate, grounding the novelty open problem. Note: the $V_{\text{present}}/V_{\text{gating}}$ split was introduced post-hoc, after this single disagreement surfaced during review; the construct was refined until it was reliably codeable, so the resulting $\kappa=0.86$ reflects iterative construct development rather than a prospective reliability study—a researcher-degrees-of-freedom caveat that applies to interpretation of that statistic.

9 Conclusion

We introduced Servo (S, G, E, V, M, π), a formal framework connecting AI Scientist systems to POMDP and BED theory, applied to four core systems, an artifact infrastructure proposal, and seven domains. The organizing hypothesis suggested by our AI-coded, human-directed survey is that calibration of the decisive validation channel *may be associated with* trustworthy, as opposed to merely mechanical, closure—a framing hypothesis that is direction-consistent under as-coded and pessimistic outcome recoding but is not adjudicated under all reasonable coding assumptions (sensitivity analysis in the supplement), and should be treated as provisional structured annotation rather than a validated empirical finding. Propositions 1–3 support only narrower claims (closure needs an identifying observation model and sufficient information; calibration governs the reward/policy channel, not posterior consistency; novelty is unidentified when no automated channel carries it), so the formalism separates observation, reward validation, and human novelty judgment rather than proving V completeness is the primary determinant.

These bottlenecks are structural—missing validator completeness, search-policy design, and memory architecture—and Servo names the vocabulary for what closing them requires.

³Coders: Claude Code (claude-opus-4-8, Anthropic), Codex CLI (gpt-5.5, OpenAI), and Antigravity (gemini-3.1-pro, Google), each run headless and model-pinned; exact invocations and per-model outputs are released in the supplement.

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Conference For AI Scientists 2026 - AI Involvement Checklist

This checklist documents the role of AI systems at each stage of this research. The checklist has two parts: a **Research Stage Assessment** (items 1–4) rating AI involvement and iteration effort, and an **AI System Documentation** section (items 5–6).

Research Stage Assessment

1. Hypothesis development.

Answer: [B]

Iteration Effort: [Medium]

Explanation: The core Servo framework tuple (S, G, E, V, M, π) , the V -completeness thesis, the H parameter, and the three open problems were conceived by the human author. A frontier LLM agent elaborated framework components, identified literature-level instantiations, and proposed theoretical connections. The human author directed all conceptual decisions and made several substantive corrections to the AI’s elaborations.

2. Experimental design and implementation.

Answer: [B]

Iteration Effort: [Medium]

Explanation: This submission contains no computational experiments. The methodology for cross-system analysis (system selection criteria, component extraction protocol, domain comparison structure) was designed by the human author. A frontier LLM agent executed literature retrieval, extracted framework components from primary sources, and performed citation-level fact verification across four verification sweeps.

3. Analysis of data and interpretation of results.

Answer: [C]

Iteration Effort: [Medium]

Explanation: The LLM agent extracted Servo components for all analyzed systems from primary sources and performed citation-level fact verification (4 verification sweeps). The human author reviewed all extractions, corrected mischaracterizations, and synthesized the cross-domain pattern. The V -completeness thesis emerged from the human author’s synthesis. Consistent with this venue’s premise, the cross-system coding is AI-conducted under human direction and disclosed as such; its rigor rests on auditability (released rubric, per-row source quotes, and regeneration/citation-validation scripts) and conservative claim-scoping rather than an independent human gold standard, which we treat as orthogonal to the framework contribution and defer to future work.

4. Writing.

Answer: [C]

Iteration Effort: [High]

Explanation: The LLM agent drafted the majority of the manuscript text through multiple revision cycles; the human author provided key conceptual passages (framework definitions in §4, core open problems in §7), directed structural decisions, reviewed all drafts, and made final editorial judgments.

AI System Documentation

5. AI systems used.

Description: A frontier large language model with code execution and long-document processing capabilities was used as the primary research and writing assistant. The system operated as an interactive CLI agent with file-system access, able to read primary sources, execute and debug Python simulations, and iteratively revise LaTeX manuscripts.

6. Observed AI Limitations.

Description: (1) Systematic citation fabrication: the agent generated plausible but non-existent quotes and statistics, requiring four dedicated citation-verification sweeps using parallel sub-agents. (2) Analytical overconfidence: the agent initially overclaimed empirical significance for mechanically-entailed theoretical consequences; multiple adversarial

review rounds were needed to correct framing and remove overclaims. (3) Context drift: over long sessions, the agent occasionally regressed previously corrected errors, requiring explicit re-verification of integrity items before each commit.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction explicitly state the three contributions (framework, cross-domain analysis, open problems). The propositions under explicit assumptions (§A) clarify the separation of observation, reward validation, and novelty judgment rather than establishing a V-completeness thesis, and the cross-domain survey is presented as illustration; the paper delimits both the assumptions and the survey’s illustrative role in §A and §8.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are discussed in §8 (framework abstracts domain cost structures, social layer, non-stationarity) and throughout the domain analysis (§6).

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [Yes]

Justification: Appendix A states three propositions, each with its assumptions (Assumption 1, Definition 1) and a proof sketch grounded in standard results on Bayesian experimental design and posterior consistency under model misspecification. The cross-domain survey is presented as illustration of these results, not as independent empirical proof.

4. Experimental result reproducibility

Question: Does the paper fully disclose all information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: This paper contains no computational experiments. The cross-domain analysis approach (system selection and component extraction from primary sources) is described in §5 and §6; machine-readable extraction sheets (core systems and the seven-domain Table 2) with a representative source quote and citation per row and citation-validating regeneration scripts, explicit inclusion/exclusion criteria, multi-vendor blind coding over 14 systems with Fleiss’ κ (plus a two-vendor Cohen’s κ and recode sensitivity for the calibration re-coding), and a counterexample search are released as supplementary material (see §8). All analyzed papers are publicly accessible. The released coding should be treated as *provisional structured annotation*: the table-generation scripts verify citation presence and non-empty source quotes but do not validate semantic entailment of each coded field; independent reproduction of the interpretive coding requires consulting the cited primary sources directly.

5. Open access to data and code

Question: Does the paper provide open access to data and code?

Answer: [Yes]

Justification: The reproducibility package is released as supplementary material in the analysis/ directory: the coding protocol (`coding_protocol.md`); coding sheets with a representative source quote and citation per row for both the core systems (`systems.csv`) and the seven-domain Table 2 (`domain_systems.csv`); the multi-vendor calibration re-coding (`multicoder/`); the reliability report (`reliability_report.md`); and the citation-validating table-regeneration scripts (`build_servo_tables.py`, `build_domain_tables.py`, `compute_calib.py`). All analyzed papers are publicly accessible; no separate dataset was generated.

6. Experimental setting/details

Question: Does the paper specify all details necessary to understand the results?

Answer: [Yes]

Justification: The cross-domain analysis methodology is described in §5 and §6. No computational experiments were performed in this submission.

7. Experiment statistical significance

Question: Does the paper report error bars or statistical significance information?

Answer: [NA]

Justification: This paper contains no computational experiments and reports no statistical results requiring error bars or significance tests.

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [NA]

Justification: This submission contains no computational experiments. The meta-analysis requires only standard document processing and no specialized compute.

9. Research Ethics

Question: Does the research conform with the NeurIPS Code of Ethics, except where CAISc’s AI authorship policies apply?

Answer: [Yes]

Justification: This is a meta-analysis of published systems with no human subjects, no proprietary data, and no safety risks. AI authorship is documented per CAISc policy.

10. Broader impacts

Question: Does the paper discuss both potential positive and negative societal impacts?

Answer: [Yes]

Justification: §8 addresses both the potential for accelerating scientific discovery and the risk of premature convergence under widespread AI Scientist deployment (particularly via the M_s underdevelopment problem and novelty non-automatability).

A Formal Analysis of Validator Completeness

We make the role of V precise by separating two objects that are easily conflated: the *observation likelihood* $p(o \mid \theta, d)$, which drives the Bayesian belief update, and the *validator* $V(o)$, a reward estimate that the policy π uses to score outcomes. The results hold in a tractable subproblem; the open-ended regime is left open.

Assumption 1 (Tractable, identifying subproblem). *(i) the search space S is bounded and the parameter space Θ is compact; (ii) T is stationary; (iii) the prior $p(\theta)$ has support containing the true θ^* ; (iv) the observation model $p(o \mid \theta, d)$ is well-specified and identifies θ (for $\theta \neq \theta'$ some reachable design separates them); and (v) the realized design sequence is sufficiently informative for posterior consistency: infinitely often it selects a reachable design whose Kullback–Leibler separation between any two surviving hypotheses is bounded below by a fixed $\delta > 0$. We do not equate this with a divergent sum of adaptive EIG terms: incremental EIG is an expected entropy reduction bounded by initial entropy in finite-state settings, so cumulative EIG can be bounded even when the posterior concentrates. The condition here is on KL-separation between hypotheses, not on accumulated EIG.*

Definition 1 (Validator as reward channel). *$V(o)$ estimates the true reward $R(s, a)$; it is calibrated if $\mathbb{E}[V(o) \mid s, a] = R(s, a)$ and biased if $\delta(s, a) = \mathbb{E}[V(o) \mid s, a] - R(s, a) \neq 0$ for some (s, a) . V is distinct from the update likelihood $p(o \mid \theta, d)$ unless explicitly coupled into it.*

Proposition 1 (Sufficiency for closure). *Under Assumption 1 the posterior concentrates on θ^* (the loop closes), and the one-step EIG rule (Equation 1) is myopically optimal for the information objective—both independently of the validator, since V is a reward channel while EIG is a functional of the prior and likelihood alone.*

Proof sketch. A well-specified, identifying likelihood with *divergent* accumulated information on a compact Θ (Assumption 1(i),(iv),(v)) is the standard sufficient condition for Bayesian posterior consistency [Ghosal and van der Vaart, 2017], giving $b_t \rightarrow \delta_{\theta^*}$. Calibration is required for neither consistency nor EIG-optimality (both depend only on the prior and likelihood); it concerns the separate reward channel (we make this precise below). The uniform-information clause (v) is essential: a sequence with $\text{EIG}(d_t) > 0$ but $\sum_t \text{EIG}(d_t) < \infty$ leaves the posterior short of a point mass even under identifiability and calibration. $\square$

Proposition 2 (Drift under misspecified coupling). *Suppose the validator enters the belief update, i.e. the effective update is $q_\theta(o) \propto p(o \mid \theta, d) \exp(\beta V(o))$ with coupling $\beta > 0$. If V depends on the outcome in a θ -dependent (non-affine) way, the posterior concentrates on the Kullback–Leibler projection $\theta^\dagger \neq \theta^*$ [Berk, 1966]: the loop closes computationally but on the wrong target. A θ -independent affine transformation of V (e.g. a constant additive bias) cancels in the log-partition and leaves θ^* unchanged; and when V is a pure policy channel ($\beta = 0$) no validator transformation moves the target—it can at most degrade design choice and slow information acquisition (interacting with Assumption 1(v)).*

Proof sketch. A coupled, misspecified update likelihood places the posterior at its KL minimizer [Berk, 1966]; the log-partition $\log \int p(o \mid \theta, d) e^{\beta V(o)} do$ relocates that minimizer iff its dependence on θ changes, which a θ -independent affine term does not. With $\beta = 0$ the update likelihood is unchanged, so θ^* is invariant to V . $\square$

Proposition 3 (Non-identifiability of novelty). *Let θ_{nov} be a novelty/significance component that is a priori independent of the identified components. If the channel likelihood of every automated validator is independent of θ_{nov} , then no automated validator admits a consistent estimator of θ_{nov} , and loop closure on novelty requires V_{human} .*

Proof sketch. If the likelihood does not depend on θ_{nov} and the prior does not couple it to identified components, the posterior marginal on θ_{nov} equals the prior for all data; θ_{nov} is unidentified and admits no consistent estimator. $\square$

The formal results support a narrower distinction: posterior closure in the tractable subproblem depends on a well-specified, identifying, sufficiently informative observation model (Proposition 1), while validator calibration governs the reward or acceptance signal used by the policy—not posterior consistency, and not which design maximizes EIG (information-optimal and reward-optimal designs coincide only when validator reward preserves the EIG ordering, and can diverge even for a perfectly calibrated V). Proposition 2 shows that drift requires an incorrectly *coupled* validator, not a mere ranking shift; Proposition 3 isolates the one component no automated validator can resolve. These results do not by themselves prove that the empirical V -completeness ordinal is necessary for all trustworthy autonomous discovery; that claim remains a design hypothesis requiring independently coded outcomes. The surveyed systems should therefore be read as motivating examples for separating observation, reward validation, and novelty judgment—not as a formal demonstration that V -completeness is the primary determinant of closure.

B Per-Domain Analysis

This appendix expands Table 2 into a per-domain reading in the Servo vocabulary, stating for each field the validator type and its reliability, the loop status that validator permits, the search policy actually deployed, and the bottleneck that follows. The coding is released as the machine-readable `domain_systems.csv`, with a representative source quote and citation keys per row; the entries below paraphrase those quotes and should be read as a structured reading of the cited work rather than an independent empirical study.

Mathematics, formal (V_{formal} ; fully closed). FunSearch [Romera-Paredes et al., 2024] and DeepSeek-Prover [Xin et al., 2024] validate against a formal or numeric oracle—an evaluator function or proof kernel returning a deterministic pass/fail. This V_{formal} is a perfect oracle in the sense of Assumption 1, so the reward channel is exactly calibrated and RL/MCTS search is stable; the loop is fully closed. The residual bottleneck is not validation but S : pretraining contamination makes it hard to certify that a discovered result is genuinely novel rather than memorized—the formal-domain instance of the novelty problem (Proposition 3).

Mathematics, natural language ($V_{\text{sem}} + V_h$; closed at low quality). Aletheia [Feng et al., 2026] produces natural-language proofs scored by LLM and human judgment. The loop closes mechanically, but V_{sem} is imperfect: of 200 audited Erdős solutions only 13 (6.5%) were rated meaningfully correct. Replacing the formal oracle with a semantic one reintroduces V_h and exposes hallucination and plagiarism as failure modes—an illustration of how a miscalibrated reward channel admits drift (Proposition 2). The Leiden Declaration [Alper et al., 2026] corroborates this from the practitioner side: AI-produced natural-language proofs may contain almost-invisible errors that resist detection without expert scrutiny, reinforcing that V_{human} is not replaceable even where automated checking exists.

Physics ($V_{\text{emp}} + V_h$; none/partial). AI Feynman [Udrescu and Tegmark, 2020] and symbolic-model discovery [Cranmer et al., 2020] fit equations to data with deterministic search. Validation is empirical and of medium reliability, but the systems carry no memory transfer across problems: each task starts fresh. The bottleneck is thus the absence of a cross-problem M , the physics instance of the semantic-memory open problem (Section 7).

Chemistry (V_{emp} surrogate; computational loop only). ChemCrow [Bran et al., 2024] and ChemReasoner [Sprueill et al., 2024] validate computationally with surrogate signals and search by beam. The computational loop closes, but physical characterization (NMR, MS) leaves the wet-lab loop open: the validator that would confirm a real discovery sits outside the automated boundary, so closure is partial at the measurement interface.

Materials science (V_{emp} via DFT; computationally closed). GNoME [Merchant et al., 2023] runs DFT-in-the-loop active learning, predicting 2.2 million candidate structures (380,000 stable), with uncertainty sampling approximating the BED ideal within the computational surrogate. DFT should be read as an approximation of physical energies rather than a fully calibrated empirical oracle for materials discovery: of the predicted stable structures, only 736 have been independently experimentally realized, and critics note that predicted compounds should not be equated with validated materials exhibiting demonstrated functionality [Cheetham and Seshadri, 2024]. Servo therefore codes this domain as computationally closed for phase-stability screening, with the physical materials-discovery loop only partly closed. The remaining bottlenecks are resource-bound: DFT wall-time and the lag before predicted structures are synthesized and characterized.

Biology ($V_{\text{emp}} + V_h$; partial). BioPlanner [O’Donoghue et al., 2023] generates experimental protocols, but the validation signals are weak and biological-validity metrics are insufficient to close the loop autonomously, so reliability is low-to-medium with effectively no search policy beyond generation. The bottleneck is V itself: without metrics that track ground-truth biological validity, the loop cannot be closed on anything trustworthy.

Social science (statistical tests; partial). Automated-social-science pipelines [Manning et al., 2024] and LLM-simulation studies [Aher et al., 2023, Park et al., 2023a] validate with statistical tests over full-factorial designs. The characteristic failure is circular validity: when the generator G and the simulated subjects share a model, the validator measures the model’s self-consistency rather than an external fact [Aher et al., 2023]—a violation of the identifying-channel condition of Assumption 1.

Computer science / algorithms (V_{emp} ; fully closed). AutoML-Zero [Real et al., 2020], algorithm discovery [Sartori and Blum, 2024], and AutoKernel [Jaber and Jaber, 2026] close the loop via reliable benchmark/empirical validation and evolutionary or LLM-guided search. Validation is high-reliability and the loop is fully closed; the residual bottleneck is interpretability—the discovered artifacts work but resist human understanding, a limitation of M and output form rather than of V .

Cross-domain reading. The bottleneck each field hits is largely a function of its V type: a perfect formal oracle yields a fully closed, contamination-limited loop; an imperfect semantic or empirical V yields hallucination, circular validity, or insufficient-metric failures; and a reliable but expensive empirical oracle (materials) shifts the bottleneck from epistemics to resources. This is the diagnostic use of Table 2: it does not claim a tested cross-domain law, but shows that the same (S, G, E, V, M, π) vocabulary localizes where each domain’s loop breaks. V type shapes whether

a domain’s loop can close *trustworthily*; the search policy π independently shapes what is *scientifically discoverable*—whether the loop reaches genuinely unexplored regions of S or remains within the already-exploited frontier. Table 2 captures the V dimension of this two-axis picture; the π dimension remains uniformly underdeveloped across all surveyed domains, with materials science the sole partial exception.